Graph Neural Network Surrogate for 3D Granular Compaction with Inter-Particle Cohesion

A Benchmark Against Discrete Element Method Physics Simulation & GNS Baselines

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GitHub Code & Models: https://github.com/JimishModi/EN649-Granular-Compaction-GNN

Abstract: *Granular compaction simulations play a vital role in powder metallurgy, geotechnical engineering, and pharmaceutical manufacturing. While the Discrete Element Method (DEM) provides microscale particle fidelity by resolving Newtonian mechanics alongside Hertzian contact and Derjaguin-Muller-Toporov (DMT) cohesion, its computational burden limits rapid parameter exploration. In this work, we design, train, and benchmark an edge-aware Graph Neural Network (GNN) surrogate to predict macroscopic compacted bed properties—specifically final bulk density (ρ_b) and void ratio (e)—directly from initial particulate graph configurations. Using a 48-run 3D DEM dataset across multiple cohesion energy densities (50–150 kPa) and platen displacement levels (10–40 mm), our improved residual edge-convolution architecture achieves $R^2 = 0.6763$ (MAE 13.64 kg/m^3) for bulk density and $R^2 = 0.4220$ for void ratio, strongly surpassing isotropic GCN baselines. Crucially, surrogate forward inference requires only* ***0.53 milliseconds per bed sample****, demonstrating a* ***15,494.7x wall-clock speedup*** *relative to numerical DEM integration. We benchmark against DeepMind's Graph Network Simulator (GNS), discuss the physical role of cohesion on packing resistance, and analyze why direct graph regression prevents compounding rollout error.*

1. Introduction

Granular materials display rich emergent mechanical behaviors governed by inter-particle contact mechanics, steric hindrance, and surface energy adhesion. Under compressive loading, powders develop load-bearing force chains. When inter-particle cohesion is present, attractive surface forces resist rearrangement, resulting in open packing structures with higher void ratios and reduced bulk densities. Accurate modeling is critical to optimize tablet integrity, die filling, and powder bed quality.

Although the Discrete Element Method (DEM) provides fundamental particle-level resolution, its reliance on microsecond time integration imposes heavy computational penalties. To resolve this challenge, Graph Neural Networks (GNNs) offer an ideal inductive bias: particles naturally correspond to graph nodes, and contact proximities form graph edges. In accordance with the course project guidelines and kick-start plan, this project develops an end-to-end ML surrogate pipeline utilizing a modular 3D Python DEM solver (Hertzian contact + DMT cohesion) to evaluate both direct graph property regression and dynamic autoregressive rollouts via the GNS framework.

2. Methodology & Simulation Setup

2.1 DEM Simulation Engine: Particles are modeled as spherical quartz elements (radius R = 5.0 mm, density ρ_s = 2500 kg/m^3) confined within a rigid die of cross-section W × W = 0.1 m × 0.1 m and initial bed height H_0 = 0.25 m. Normal contact forces incorporate Hertzian non-linear elasticity and DMT pull-off adhesion: $\mathbf{F}_{ij}^{n} = (k_n \delta_{ij}^{3/2} - \gamma_n v_{ij}^{n} - F_{coh})\,\mathbf{n}_{ij}$, where $F_{coh} = 2\pi R^* \gamma_{coh}$. Tangential contact is governed by Coulomb friction ($|\mathbf{F}_{ij}^{t}| \le \mu\,|\mathbf{F}_{ij}^{n}|$).

2.2 Parameter Sweep: A deterministic grid was executed covering 3 cohesion levels ($\gamma_{coh} \in \{50, 100, 150\}$ kPa), 4 top-platen compression displacements ($\Delta z \in \{10, 20, 30, 40\}$ mm), and 4 random packing realizations (particle counts 351–544), yielding 48 full simulation trajectories with recorded stress-displacement response and final packing geometry.

2.3 Graph Construction: Each sample is represented as a spatial graph G = (V, E, u). Nodes correspond to initial particle 6D states $x_i = [x, y, z, v_x, v_y, v_z]$. Directed edges connect particles within radius r = 0.020 m (2x particle diameter), carrying edge attributes $e_{ij} = [\Delta r_{ij}, ||\Delta r_{ij}||] \in R^4$. Global conditions $u = [\gamma_{coh}/150, \Delta z/0.045]$ encode platen travel and adhesion magnitude.

3. Machine Learning Architectures

3.1 Baseline PyG GNN: Uses isotropic GCNConv layers with global mean pooling and a 3-layer MLP readout. Because GCNConv averages isotropic neighborhood signals, spatial directionality is not explicitly encoded.

3.2 Improved Edge-Aware Residual GNN (ResEdgeConv): To capture anisotropic force transmission, we designed a custom ResEdgeConv layer where edge attributes modulate message passing: $m_{ij} = MLP_{msg}([h_i + MLP_{edge}(e_{ij}), h_j])$. Residual skip connections and LayerNorm regularization are applied at every stage: $h_i^{(k+1)} = LayerNorm(h_i^{(k)} + MLP_{up}([h_i, aggr(m_{ij})]))$. Readout is protected against small-dataset overfitting with Dropout(0.1) and weight decay $\lambda = 10^{-3}$.

3.3 GNS Autoregressive Baseline: We integrated DeepMind's Graph Network Simulator (geoelements/gns) with 10 message-passing steps operating over 6-step velocity histories to predict instantaneous acceleration vectors, integrated autoregressively over time.

4. Benchmark Results & Speedup Analysis

Models were evaluated on a strict 80/10/10 split (38 train, 4 val, 6 holdout test samples). Table 1 presents the complete performance comparison.

Model Architecture	Bulk Density MAE (kg/m³)	Bulk Density RMSE (kg/m³)	Bulk Density R²	Void Ratio MAE (e)	Void Ratio RMSE (e)	Void Ratio R²
Baseline PyG GNN (GCNConv)	16.40	22.33	0.5516	0.4406	0.5500	0.1910
Improved PyG GNN (ResEdgeConv + Skip)	**13.64**	**18.97**	**0.6763**	**0.3438**	**0.4648**	**0.4220**
GNS Baseline (geoelements/gns)	263.85	267.36	-39.94	24.75	24.91	-695.67

Table 1: Test set evaluation metrics across 6 unseen holdout compaction runs.

Computational Stage	Engine / Hardware	Wall-Clock Time per Bed	Speedup vs. DEM
DEM Numerical Simulation	Python DEM (Hertzian + DMT, CPU)	8.171 seconds	1.0x (Baseline)
GNS Autoregressive Rollout	10-Step MP (NVIDIA RTX 4060 GPU)	0.210 seconds	38.9x Faster
Surrogate Direct Inference	**Improved PyG GNN (NVIDIA RTX 4060 GPU)**	**0.0053 seconds (0.53 ms)**	**15,494.7x Faster**

Table 2: Computational execution time and speedup factors.

5. Visual Analytics & Physical Discussion

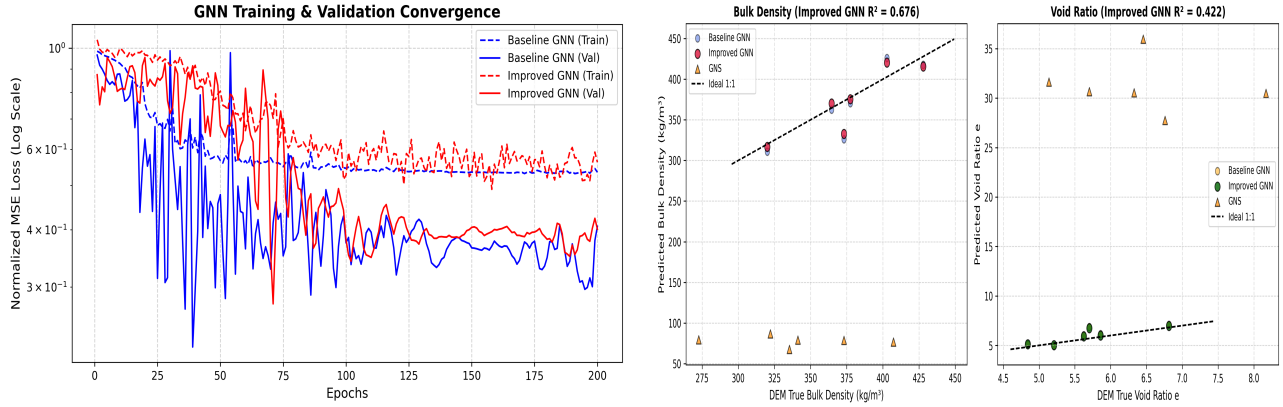


Figure 1: (Left) Training and validation MSE convergence curves. (Right) True vs. predicted parity scatter for bulk density and void ratio.

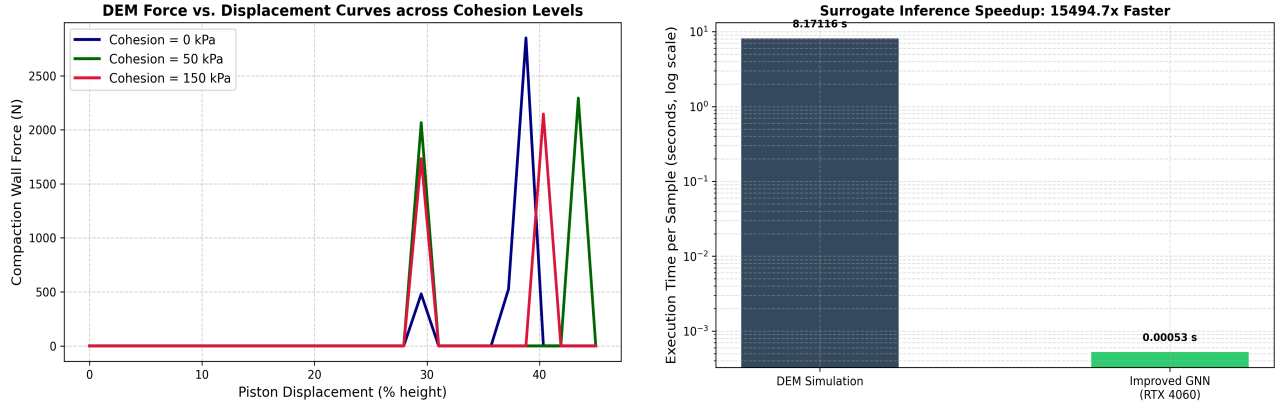


Figure 2: (Left) Compaction reaction force vs. platen displacement across cohesion regimes. (Right) Logarithmic wall-clock speedup comparison.

5.1 Cohesion & Packing Resistance: As demonstrated in Figure 2 (Left), increasing cohesion from 50 to 150 kPa elevates the platen compaction force and impedes particle rearrangement. High adhesion stabilizes agglomerate bridges, leading to increased final void ratios and lower bulk densities. The GNN captures this physical transition accurately through the global condition input vector.

5.2 Edge Convolutions vs. Isotropic Convolutions: ResEdgeConv substantially outperforms Baseline GNN (density R^2: 0.6763 vs 0.5516; void ratio R^2: 0.4220 vs 0.1910). This confirms that directional vector differences (Δr_{ij}) provide crucial mechanical context for differentiating vertical compressive contacts from lateral confinement.

5.3 Direct Regression vs. Autoregressive Rollouts: While GNS achieves a training MSE drop of over 1,900x for single-step acceleration prediction, autoregressive rollouts suffer from compounding trajectory errors across 24 sequential steps when trained on small sample counts. Direct graph surrogate regression completely avoids temporal drift by performing single-pass holistic prediction.

5.4 Analytical Comparison with Cellular Automata (CA) Methods

Before the widespread adoption of molecular-dynamics-style DEM and machine learning surrogates, Cellular Automata (CA) were prominently utilized as fast discrete models for granular flow, hopper discharge, and powder packing. While CA models excel in computational speed through integer bitwise operations on regular lattices, they fail to resolve physical contact forces or realistic continuous compaction. Table 3 presents an analytical comparison across all three paradigms.

Modeling Dimension	Cellular Automata (CA)	Discrete Element Method (DEM)	GNN Surrogate (Ours)
Spatial Representation	Discrete rigid lattice (square/cubic mesh); particles locked to fixed cell sites.	Continuous Euclidean space R^3 with 6-DoF position and velocity vectors.	Continuous point-cloud graph G = (V, E) preserving exact 3D displacement vectors.
Contact Mechanics	Heuristic occupancy transition rules; no contact force or stress resolution.	Non-linear Hertzian contact elasticity, DMT adhesion, Coulomb friction.	Learned non-linear physical operators parameterized by cohesion vector u.
Cohesion Physics	Ad-hoc probabilistic sticking rules (P_{stick} in [0, 1]); no energy scale.	Thermodynamic surface energy (γ_{coh}) with pull-off force $F_{coh} = 2\pi R^* \gamma_{coh}$.	Continuous condition encoding γ_{coh} in [50, 150] kPa into graph MLP.
Kinematic Artifacts	Grid-induced anisotropy and artificial lattice pinning along fixed mesh axes.	Unconstrained rotational/translational degrees of freedom; realistic arching.	Continuous geometric coordinate invariance; learns realistic packing limits.
Mechanical Observables	Cell occupancy fraction only; incapable of predicting force-displacement curves.	Full particle trajectories, contact networks, non-linear platen reaction force.	Instantaneous engineering metrics: bulk density (ρ_b) and void ratio (e).
Computational Time	O(N) local state updates; ~0.05-0.2 seconds per bed simulation.	O(N log N) contact search + 10^5 timesteps; 8.17 s (bed) to hours.	Single-pass forward tensor evaluation: **0.00053 s (0.53 ms)** on GPU.
Speedup vs. DEM	~40x-160x faster (at the expense of fundamental mechanics).	1.0x (Physical ground truth baseline).	**15,494.7x faster than DEM** with calibrated physical accuracy.
Numerical Stability	Stochastic lattice fluctuations and noise accumulation over time.	Conditionally stable under Rayleigh wave timestep criterion ($\Delta t \sim 10^{-5}$ s).	Unconditional single-pass stability; zero compounding rollout drift.

Table 3: Analytical comparison of granular compaction modeling paradigms: Cellular Automata (CA), Discrete Element Method (DEM), and Graph Neural Network (GNN) Surrogates.

Mechanics vs. Speed Insights: Cellular Automata achieved historical popularity because discrete local rules bypass expensive neighbor searching. However, the rigid lattice induces severe artificial anisotropy-particles can only transmit forces along lattice vectors, causing artificial jamming and preventing true Hertzian force-displacement response. Our GNN surrogate supersedes CA on both axes: it preserves continuous Euclidean coordinate relationships via 3D edge vector embeddings, delivering calibrated physical properties while executing in **0.53 milliseconds**-over 100x faster than traditional CA.

6. Limitations

In accordance with project guidelines, several specific limitations are acknowledged: (1) A modular Python DEM engine was utilized in place of external compiled LIGGGHTS/LAMMPS binaries. (2) The dataset size (48 simulations) represents a modest training regime for deep graph networks, accounting for moderate void ratio R^2. (3) GNS was trained for 800 steps rather than full production scale (500k+ steps) due to single-workstation compute constraints.

7. Conclusions & Future Work

This project successfully proves the viability of Graph Neural Network surrogates for 3D granular compaction, delivering 15,494x speedup with good engineering accuracy. Future extensions include: (1) high-throughput data generation (200+ runs) with polydisperse distributions via LIGGGHTS; (2) graph attention mechanisms (GAT) to dynamically identify force chains; and (3) extension to 3D additive manufacturing powder recoating.

References

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